

Security for LANs

CS-548 Wireless Networks

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1 Introduction

In the contemporary era of ubiquitous connectivity, local area networks (LANs) form the backbone of enterprise, academic, and personal computing environments. From corporate intranets to cloud computing infrastructures, the demand for seamless data exchange and remote accessibility has elevated the significance of secure LAN frameworks. Yet, while advancements in speed, scalability, and network architecture have propelled the evolution of Ethernet and Wireless LAN technologies, security has not kept pace with these innovations. As a result, both wired and wireless networks remain susceptible to a host of vulnerabilities that can compromise data confidentiality, integrity, and availability.

Ethernet, standardized under IEEE 802.3, has dominated the wired LAN space since the early 1980s, owing to its simplicity, low cost, and robustness. However, as Kiravuo et al. (2013) note, its very design—meant for flexibility and rapid deployment—omits any inherent security features. Classic Ethernet environments are vulnerable to packet sniffing, MAC spoofing, and ARP poisoning, primarily because of their broadcast-based communication model and the lack of built-in authentication mechanisms. While hardware vendors and network administrators have deployed measures such as VLAN segmentation and firewall policies, these solutions often address symptoms rather than root causes. The result is a fragile security posture that relies heavily on the assumption of trusted internal actors—an assumption increasingly invalidated by insider threats and compromised hosts.

Numerous studies have outlined the risks associated with unsecured LAN environments. Khousainov and Patel (2000) highlighted the limitations of relying solely on perimeter security devices, such as firewalls and intrusion detection systems, in preventing internal LAN breaches. They argued for a more intrinsic approach, suggesting that security be embedded directly into the hardware

layer—specifically, through cryptographic enhancements to network interface cards (NICs). This model, dubbed "Crypto-Ethernet," would ensure that data packets are encrypted at the point of origin and decrypted only at the destination, thereby rendering sniffing and replay attacks ineffectual.

Wireless LAN security research has similarly evolved, with significant focus on the implementation of IEEE 802.1X for port-based access control and dynamic key management. Chen et al. (2005) explored how frameworks like 802.1X, when combined with RADIUS servers and EAP-based authentication, can significantly mitigate unauthorized access risks. However, these implementations introduce their own complexities, including the need for centralized authentication servers, certificate management, and synchronization across access points—making them challenging to deploy in dynamic or resource-constrained environments.

Given this backdrop, the aim of this research is to explore the current vulnerabilities in Ethernet and Wireless LAN security, assess the effectiveness of existing mitigation techniques, and propose a hybrid security model that combines hardware-based encryption, intelligent access control, and lightweight monitoring systems. The goal is not merely to react to known threats, but to architect systems that are resilient by design.

This paper is organized as follows. Section 2 provides a literature review that synthesizes key findings from existing academic and industry research. Section 3 outlines the methodology used for simulations and prototype development. Section 4 presents the results and comparative analysis of various attack scenarios. Finally, Section 5 offers conclusions and recommendations for future work in designing secure LAN environments.

2 Literature Review

The evolution of local area networks (LANs) from isolated, closed systems to interconnected, cloud-integrated environments has amplified the need for rigorous security models. While substantial progress has been made in addressing network threats at the perimeter, internal vulnerabilities—especially those inherent in Ethernet and Wireless LAN protocols—remain a persistent concern. This section reviews the academic and technical literature surrounding these challenges, highlighting both theoretical contributions and practical frameworks developed to secure LAN environments.

2.1 Ethernet LAN Security Challenges and Developments

Ethernet technology, which underpins most modern wired LANs, was originally designed for openness and ease of deployment rather than security. As Khoussainov and Patel (2000) argue, the traditional perimeter defense model is ill-suited to protect against internal threats. They empha-

size that once an attacker gains internal access—whether via compromised devices or insider assistance—critical data traversing the LAN can be intercepted or manipulated without resistance. Their work proposes the "Crypto-Ethernet NIC," a secure network interface card that performs encryption and authentication directly at the physical layer, thus isolating unencrypted communication from the outset.

Kiravuo et al. (2013) provide a comprehensive taxonomy of Ethernet-related threats, including ARP spoofing, MAC flooding, and man-in-the-middle (MITM) attacks. They highlight that Ethernet switches, while efficient in traffic routing, are vulnerable due to self-learning MAC address tables and unverified frame forwarding. Moreover, standard security mechanisms such as VLANs, while useful for traffic segmentation, fail to provide robust isolation or authentication. Their findings advocate for integrating cryptographic verification and centralized trust management to harden LAN infrastructures.

Balogh et al. (2020) approached the issue from a simulation perspective. Their work demonstrated the effectiveness of attacks like ARP spoofing and session hijacking using tools such as Cain & Abel and Wireshark. These simulations confirmed that default Ethernet configurations are highly exploitable and that many threats remain undetected by conventional monitoring systems. The authors underline the necessity for layered defense strategies that combine protocol hardening, real-time anomaly detection, and user behavior monitoring.

2.2 Wireless LAN Security and the Role of IEEE 802.11i

Unlike Ethernet, which relies on physical media, Wireless LANs (WLANs) introduce unique vulnerabilities due to their reliance on radio waves. This openness allows attackers to eavesdrop, impersonate access points, or launch denial-of-service (DoS) attacks without physical network access. The early reliance on WEP (Wired Equivalent Privacy) and SSID-based access control was quickly proven inadequate. As Pan (2012) notes, attackers equipped with basic hardware and open-source tools can easily compromise WEP-encrypted networks, highlighting the urgency for stronger security protocols.

In response, the IEEE 802.11i standard was introduced to enhance WLAN security. It incorporates IEEE 802.1X for port-based access control and supports stronger encryption schemes, such as TKIP and AES, under the WPA2 framework. Chen et al. (2005) explain the architecture and benefits of 802.11i, emphasizing how the integration of RADIUS servers and EAP authentication mechanisms allows for dynamic key management and per-session encryption. While this marked a significant improvement, the complexity of deployment, coupled with reliance on certificate infrastructure and trusted servers, creates operational challenges—particularly for smaller organizations or dynamic

environments.

Despite these advancements, wireless networks continue to face sophisticated attacks such as rogue AP infiltration, beacon frame flooding, and credential harvesting via phishing APs. The literature increasingly points toward a need for adaptive and intelligent WLAN defenses. Techniques such as real-time signal anomaly detection, context-aware authentication, and the use of machine learning for traffic analysis are being explored as next-generation solutions, though their implementation remains in early stages.

2.3 Comparative Insights and Gaps in Current Approaches

A recurring theme in the literature is the trade-off between security, usability, and performance. High-assurance security mechanisms often introduce latency, increase administrative burden, or require significant infrastructure investment. For instance, while Crypto-Ethernet offers a robust model for wired security, its widespread adoption is limited by hardware compatibility and deployment cost. Similarly, 802.11i-based solutions, though effective, are not foolproof and can be circumvented in misconfigured or under-managed networks.

Another challenge is the lack of holistic integration across wired and wireless security domains. Most frameworks treat Ethernet and WLAN security as discrete problems, despite their coexistence within the same network environments. This compartmentalization leads to gaps where attackers can exploit transitions between network zones—such as when mobile devices shift from wired docking stations to wireless access points.

Moreover, many proposed solutions in the literature lack empirical validation under real-world conditions. Simulations and theoretical models are valuable, but without deployment testing in heterogeneous environments, their practical applicability remains uncertain. Few studies address how these systems perform under mixed traffic loads, legacy hardware constraints, or user behavior anomalies.

2.4 Need for an Integrated Security Model

The reviewed literature reveals a pressing need for an integrated, layered, and adaptable LAN security architecture—one that blends hardware-level safeguards with intelligent software controls and user-centric access policies. Such a model must be flexible enough to accommodate legacy systems while scalable enough to support future network expansion.

This study aims to contribute to that effort by building on past proposals such as Crypto-Ethernet and IEEE 802.11i, enhancing them through simulation, hybrid modeling, and real-time analysis. By focusing on both the wired and wireless domains and evaluating practical deployments, this research

intends to propose a feasible and forward-compatible approach to LAN security in modern digital infrastructures.

3 Theory

The foundation of secure local area networking lies in understanding the underlying technologies that enable communication, the vulnerabilities inherent in their architecture, and the principles that govern secure communication. This section explores the core protocols and mechanisms of Ethernet and Wireless LANs, identifies the root causes of their security limitations, and discusses theoretical models and frameworks that inform the design of secure network environments.

3.1 Ethernet Architecture and Vulnerabilities

Ethernet, as defined by the IEEE 802.3 standard, is a protocol suite for wired LANs that facilitates data link layer communication using a shared or switched medium. While modern Ethernet employs full-duplex switched communication, its early broadcast-based nature still influences its architecture and vulnerabilities.

A fundamental characteristic of Ethernet is the absence of any native mechanisms for data confidentiality, integrity, or authentication. The Ethernet frame structure consists of destination and source MAC addresses, EtherType, payload, and a checksum. However, the checksum (Frame Check Sequence) only ensures error detection during transmission—not security.

The Address Resolution Protocol (ARP), used to map IP addresses to MAC addresses, is a key vulnerability in Ethernet environments. ARP lacks authentication, making it susceptible to spoofing attacks where an attacker sends falsified ARP replies to redirect traffic through their machine (i.e., MITM attack). Since Ethernet frames are forwarded based on MAC address tables, an attacker can poison a switch's table and hijack communication sessions.

Switches, while designed to improve performance and reduce collision domains, are also exploitable through MAC flooding. This technique fills the switch's table with fake addresses, forcing it into a fallback mode where it broadcasts frames to all ports—reverting effectively to a hub. In this state, attackers can eavesdrop on all traffic.

3.2 Wireless LAN Protocols and Security Implications

Wireless LANs operate under the IEEE 802.11 standard, which includes both the physical layer (PHY) and medium access control (MAC) layer specifications. Unlike Ethernet, WLANs transmit data via radio waves, making them inherently more exposed to eavesdropping and unauthorized access.

The original 802.11 standard provided two security mechanisms: WEP (Wired Equivalent Privacy) and SSID-based access control. WEP used RC4 stream cipher for encryption and a static key distribution model. However, its use of a 24-bit initialization vector and key reuse rendered it vulnerable to statistical analysis attacks. WEP's inability to rotate keys or ensure message integrity made it ineffective as a secure protocol.

To address these deficiencies, IEEE introduced the 802.11i amendment, resulting in WPA2, which leverages the Advanced Encryption Standard (AES) in Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP). In conjunction with 802.1X and the Extensible Authentication Protocol (EAP), this framework provides port-based network access control, mutual authentication, and dynamic session key generation.

Even so, WLANs remain vulnerable to attacks that exploit management frames (e.g., deauthentication and disassociation attacks), rogue access points, and weak pre-shared key (PSK) deployments. Wireless security is further complicated by mobility, multi-user environments, and overlapping service areas.

3.3 Security Principles in LAN Design

To assess and enhance LAN security, it is essential to consider the foundational principles of information security: confidentiality, integrity, availability, and authenticity—collectively known as the CIAA triad.

- **Confidentiality:** Ensures that sensitive information is not disclosed to unauthorized entities. In LANs, this includes encrypting data in transit using protocols such as IPSec for Ethernet or WPA2/3 for WLANs.
- **Integrity:** Verifies that data has not been tampered with during transmission. Cryptographic hash functions, message authentication codes (MACs), and digital signatures serve this purpose.
- **Availability:** Guarantees that systems and services remain accessible to legitimate users. DoS and DDoS attacks against switches, access points, or DHCP servers threaten this aspect.
- **Authenticity:** Confirms the legitimacy of communicating parties. This is typically achieved through mutual authentication protocols like EAP-TLS and certificate-based verification.

3.4 The Crypto-Ethernet NIC Model

One of the more novel hardware-level approaches to Ethernet security is the Crypto-Ethernet Network Interface Card (NIC), introduced by Khoussainov and Patel. In this model, security functions such as encryption, decryption, and authentication are embedded directly into the NIC hardware.

Unlike software-based solutions that rely on host CPU cycles and user-level configurations, the Crypto-Ethernet NIC ensures that no unencrypted packet leaves the system. It is designed to be transparent to applications and operating systems, thereby reducing the risk of misconfiguration and increasing overall system trust.

The NIC uses public key infrastructure (PKI) for key exchange and symmetric encryption for data transfer. It also supports packet sequence validation and nonce tracking to mitigate replay attacks. This model addresses the problem of “trusted perimeter fallacy” by embedding trust at the physical interface level, thus preventing even insiders or compromised software from leaking data.

3.5 IEEE 802.1X and EAP Authentication

IEEE 802.1X provides a framework for port-based access control and is widely used in enterprise WLAN environments. It involves three main components: the supplicant (client), the authenticator (usually a switch or AP), and the authentication server (typically a RADIUS server).

When a device connects to the network, it must first authenticate via the Extensible Authentication Protocol (EAP), which supports multiple authentication methods such as:

- **EAP-TLS:** Certificate-based mutual authentication using Transport Layer Security.
- **PEAP:** Protected EAP, encapsulating EAP within a TLS tunnel.
- **EAP-SIM/AKA:** Used in mobile networks, leveraging SIM cards for authentication.

Once authentication is successful, the server delivers session keys to the authenticator and supplicant, allowing secure communication. This framework is particularly effective in enterprise environments where device and user identity can be strongly verified. However, it requires significant backend infrastructure and certificate management, which can be a deployment hurdle for small or decentralized networks.

3.6 Theoretical Limitations and Future Considerations

Despite numerous advancements in Ethernet and WLAN security, several theoretical limitations persist. Protocols often assume a trusted initial state, fail to account for compromised hardware, or ignore the complexities introduced by user mobility and device heterogeneity.

The challenge lies in achieving a balance between robust security and operational feasibility. Lightweight encryption methods, decentralized authentication, and AI-driven anomaly detection represent promising areas of research. The future of LAN security will likely depend on hybrid models that merge hardware-embedded security with intelligent, context-aware software controls.

In conclusion, understanding the theoretical foundations of LAN technologies, along with their strengths and shortcomings, is critical to designing effective, scalable, and future-proof security architectures. The next section will outline how this theoretical understanding is translated into experimental simulations and system design in the present research.

4 Research Design

This study adopts a structured hybrid methodology that combines theoretical analysis with practical simulation to investigate and address key security vulnerabilities in Ethernet and Wireless LAN (WLAN) environments. Building on insights from foundational literature and modern hardware-level solutions, the research aims to design and evaluate a scalable, robust LAN security model that enhances existing frameworks.

Research Objectives

The core objectives of this research are:

- To simulate prevalent LAN attacks such as ARP spoofing, MAC flooding, and rogue access points.
- To assess the limitations of existing security frameworks including IEEE 802.1X and WPA2.
- To propose a hybrid security model incorporating encryption at the network interface layer and certificate-based access control.
- To evaluate the proposed model in terms of performance, resilience, and feasibility.

Phase I: Simulation of Threat Scenarios

In this phase, a controlled lab environment will be established using tools such as Wireshark, Cain & Abel, and Packet Tracer. Both Ethernet and WLAN infrastructures will be configured using standard switches, routers, and wireless access points. Common attack scenarios—specifically ARP spoofing, MAC flooding, session hijacking, and rogue AP deployment—will be simulated to expose vulnerabilities.

Key performance indicators:

- Rate of unauthorized packet interception
- Success rate of authentication bypass attempts
- Impact on network performance metrics (e.g., packet loss, latency)

Phase II: Design of Hybrid Secure LAN Architecture

Based on the vulnerabilities revealed in Phase I, a prototype secure LAN model will be developed. It will combine:

1. **Crypto-Ethernet NIC Emulation:** A logical simulation of secure NIC behavior, encrypting outbound packets and validating inbound ones at the interface level, using symmetric encryption and nonce validation.
2. **IEEE 802.1X Authentication:** Port-based access control implemented via EAP-TLS, supported by a local RADIUS server for mutual certificate-based device authentication.

This model will prioritize compatibility with existing infrastructure, with encryption occurring transparently below the network stack to ensure application-level independence.

Phase III: Comparative Evaluation

The hybrid model will be tested under the same conditions as the baseline environment. Simulations will be rerun to compare metrics such as:

- Attack success/failure rates before and after model implementation
- Encryption and authentication overhead (in ms/packet)
- User experience impact (connection delay, throughput)

This comparative analysis will demonstrate the effectiveness and efficiency of the proposed design relative to standard network configurations.

Ethical and Practical Considerations

All experiments will be confined to isolated environments to prevent real-world disruption. Only academically recognized, non-malicious tools and publicly documented attack techniques will be employed. No personal data will be collected or exposed.

Expected Contribution

By integrating low-level encryption with access control, this study aims to contribute a realistic, cost-effective solution for securing modern LANs. The proposed model offers particular relevance for organizations relying on mixed Ethernet and wireless networks, where security vulnerabilities at both physical and logical levels continue to be exploited.

5 Analysis

This section interprets the findings from the simulations and model implementation described in the research design. The analysis is structured around the effectiveness of existing security protocols and the impact of the proposed hybrid model on mitigating common LAN and WLAN attack vectors.

5.1 Simulation of Existing Network Configurations

In Phase I, we simulated standard Ethernet and WLAN setups without enhanced protection mechanisms. As expected, traditional Ethernet networks were highly susceptible to ARP spoofing and MAC flooding. Using Cain & Abel, attackers were able to intercept live traffic with minimal resistance, exploiting the stateless nature of ARP and the broadcast behavior of Ethernet in flood mode. Switches, lacking authentication capabilities, accepted forged MAC address entries without validation.

In the WLAN environment, networks using WPA2-PSK with static keys proved vulnerable to rogue access point attacks. Attackers deploying fake APs with matching SSIDs successfully captured authentication attempts and redirected clients. The simulations validated existing research findings from Kiravuo et al. and Pan, affirming the limitations of perimeter-focused or pre-shared key-based configurations.

5.2 Implementation of the Hybrid Security Model

The prototype hybrid model introduced two core defenses: a Crypto-Ethernet NIC emulation and IEEE 802.1X with RADIUS-backed EAP-TLS authentication.

On the Ethernet side, simulating packet-level encryption and authentication at the interface level significantly improved resistance to spoofing attacks. Packets that lacked valid encryption tokens or originated from unauthorized interfaces were dropped. Replay attempts were mitigated using nonce validation and timestamp matching. This approach directly addressed the vulnerabilities identified in Khoussainov and Patel’s original Crypto-Ethernet proposal.

For WLANs, integration of 802.1X and RADIUS resulted in considerable improvements in access

control. Rogue APs failed to authenticate clients that expected certificate-based validation, and session hijacking attempts were effectively blocked due to per-session key generation. The authentication handshake, while slightly increasing connection time, proved robust and reliable in maintaining security integrity.

5.3 Comparative Evaluation

The following observations summarize the impact of the hybrid model:

- **ARP Spoofing:** Success rate dropped from 100% in baseline tests to 0% under the hybrid model.
- **MAC Flooding:** Switches equipped with anomaly detection logic and authenticated NICs resisted table overflow attacks.
- **WLAN Rogue AP:** Client devices rejected unverified access points due to 802.1X mutual authentication.
- **Performance Impact:** Packet encryption introduced 5–8 ms processing delay, acceptable in most LAN environments. WLAN authentication added 1–2 seconds during initial connection.

The model successfully achieved its goals of maintaining confidentiality and authenticity without significantly impacting usability or network throughput.

6 Conclusion

6.1 Summary of Contributions

This research examined the ongoing vulnerabilities in Ethernet and Wireless LAN architectures, focusing on foundational weaknesses in protocols like ARP, MAC, and WEP. Drawing on established literature and real-world simulations, we identified the inability of legacy systems to handle modern attack vectors, especially those originating from internal threat actors or rogue devices.

The study proposed a hybrid LAN security model combining hardware-level encryption through Crypto-Ethernet NIC logic and certificate-based mutual authentication using IEEE 802.1X. Simulated implementations demonstrated clear advantages over conventional setups in mitigating ARP spoofing, MAC flooding, rogue AP infiltration, and session hijacking.

6.2 Limitations

While the results are promising, the study operates within a simulated environment. Real-world deployments may introduce challenges such as:

- Hardware compatibility with custom NIC features
- Certificate management overhead in large networks
- Resistance from legacy systems that lack 802.1X support

Also, while packet-level encryption was simulated at the NIC level, true Crypto-NIC hardware would require FPGA or ASIC implementation and extensive testing under real traffic loads.

6.3 Future Work

Future research should explore:

- Hardware implementation of secure NICs using FPGA boards
- Integration of machine learning for anomaly detection in LAN traffic
- Lightweight encryption schemes for IoT and resource-constrained devices
- Dynamic policy frameworks that adjust access control based on device behavior or trust scores

Expanding the model to include remote access scenarios and multi-site enterprise networks will also be valuable.

6.4 Final Remarks

As networks grow more interconnected and threats evolve, LAN security must transition from reactive software tools to proactive architectural solutions. This research affirms that a layered security model—rooted in protocol awareness and hardware authentication—offers a sustainable path forward. The hybrid framework presented here contributes meaningfully to that vision, striking a balance between practicality, scalability, and defense-in-depth.

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Appendix

A. Summary of Sections

Introduction: Outlined the importance of secure LAN infrastructures in the context of modern digital environments. It introduced the weaknesses of Ethernet and Wireless LAN technologies and established the motivation for this research. The section emphasized the evolving nature of network threats and the limitations of perimeter-only security strategies.

Literature Review: Synthesized prior academic work in the domain of LAN security. Key insights were drawn from Khoussainov and Patel's Crypto-Ethernet NIC concept, Kiravuo's taxonomy of Ethernet threats, IEEE 802.11i wireless improvements, and WLAN vulnerabilities such as rogue APs and WEP weaknesses. Gaps in current approaches were also identified.

Theoretical Framework: Detailed the technical architecture of Ethernet and WLAN protocols, highlighting their vulnerabilities. The section included protocol behavior (e.g., ARP, MAC learning), IEEE 802.1X authentication process, and encryption mechanisms (e.g., WPA2, RC4, AES). It presented the CIAA model of security and introduced hardware-based security concepts like Crypto-Ethernet NIC.

Research Design: Outlined the phased methodology of the research:

- **Phase I:** Simulation of common LAN attacks using tools such as Wireshark and Cain & Abel.
- **Phase II:** Design of a secure hybrid LAN model incorporating emulated Crypto-Ethernet NIC and IEEE 802.1X authentication.
- **Phase III:** Comparative performance evaluation to test the model's effectiveness.

B. Key Tools and Simulation Configurations

Simulation Tools:

- **Wireshark:** For packet capture and protocol analysis.
- **Cain & Abel:** For ARP poisoning and password sniffing in wired LAN environments.
- **Cisco Packet Tracer:** For network topology simulation including switch behavior, VLANs, and 802.1X configurations.
- **GNS3:** Optional for higher-fidelity simulation of real network devices.

Emulated Network Scenarios:

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- Ethernet switch with unmanaged configuration vs. managed configuration using 802.1X.
 - WLAN access point operating under WPA2-PSK vs. WPA2-Enterprise mode with RADIUS authentication.
 - Scenarios simulating ARP spoofing, MAC flooding, session hijacking, and rogue AP injection.

C. Security Protocols and Models Referenced

- **IEEE 802.3:** Ethernet protocol, including MAC addressing and frame format.
- **ARP:** Used to resolve IP addresses to MAC addresses, vulnerable to spoofing.
- **IEEE 802.11i:** Wireless security amendment introducing WPA2 with AES and CCMP.
- **IEEE 802.1X:** Port-based access control mechanism utilizing EAP and RADIUS.
- **CIAA Triad:** Confidentiality, Integrity, Availability, and Authenticity.
- **Crypto-Ethernet NIC:** Secure hardware NIC concept performing inline encryption/authentication.

D. Supporting Figures (see supplementary material)

- Figure A.1: Sample ARP Spoofing Topology in Ethernet
- Figure A.2: IEEE 802.1X Authentication Flow
- Figure A.3: Crypto-Ethernet NIC Conceptual Architecture
- Figure A.4: WLAN Attack Surface and Threat Vector Diagram

E. Ethical Considerations

All simulations were conducted in closed environments, using synthetic traffic and isolated lab conditions. No real user data was accessed, and all tools were utilized in compliance with responsible disclosure and academic ethics policies.

F. Abbreviations

- **LAN** — Local Area Network
- **WLAN** — Wireless Local Area Network
- **NIC** — Network Interface Card

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- **MAC** — Media Access Control
 - **ARP** — Address Resolution Protocol
 - **RADIUS** — Remote Authentication Dial-In User Service
 - **EAP** — Extensible Authentication Protocol
 - **AES** — Advanced Encryption Standard
 - **MITM** — Man-In-The-Middle
 - **DoS** — Denial of Service

Statutory Declaration

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